Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:
I. If a patient has Fever AND Cough → Possible Flu
Ii. If a patient has Fever AND Rash → Possible Measles
Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia
Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True
(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 1 – Smart Medical Diagnosis System

(a) Propositional Logic representation

The hospital uses a rule-based Logical Agent to identify possible diagnoses from symptoms. Propositional symbols represent statements that are either true or false.

Symbol	**Meaning**
F	Patient A has Fever
C	Patient A has Cough
R	Patient A has Rash
B	Patient A has Breathlessness
FL	Patient A has Possible Flu
M	Patient A has Possible Measles
P	Patient A has Possible Pneumonia

Rule 1: Fever AND Cough imply Possible Flu.

$$\mathbf{(F \land C) \rightarrow FL}$$

Rule 2: Fever AND Rash imply Possible Measles.

$$\mathbf{(F \land R) \rightarrow M}$$

Rule 3: Cough AND Breathlessness imply Possible Pneumonia.

$$\mathbf{(C \land B) \rightarrow P}$$

Given facts:

F = True, C = True, B = True

Here, ∧ means AND and → means "implies" or "if–then". Rash is not supplied as a true fact, so Possible Measles cannot be derived from the given information.

(b) Forward Chaining

Forward Chaining is a data-driven inference method. It starts from known facts and applies any rule whose conditions are satisfied, adding the resulting conclusion to the knowledge base.

Step	Rule / check	Inference and result
1	Initial facts	F, C, and B are true.
2	(F ∧ C) → FL	F and C are true; derive FL (Possible Flu).
3	(F ∧ R) → M	R is not known to be true; do not derive M.
4	(C ∧ B) → P	C and B are true; derive P (Possible Pneumonia).
5	Check rules again	No additional diagnosis can be derived; stop.

Forward Chaining result: Possible Flu and Possible Pneumonia are derived. Possible Measles is not derived because Rash is not established as true.

(c) Backward Chaining to verify Pneumonia

Backward Chaining is goal-driven. It begins with the target conclusion and works backward to check whether the conditions required by a rule are supported by facts.

Goal: P (Patient A has Possible Pneumonia). The relevant rule is (C ∧ B) → P. Therefore, the system must prove both C and B.

Goal / subgoal	Evidence	Status
P: Possible Pneumonia	Requires C ∧ B under Rule 3	To be proved
C: Cough	Given fact C = True	Proved
B: Breathlessness	Given fact B = True	Proved
P: Possible Pneumonia	Both subgoals are true; apply Rule 3	Goal proved

Goal reduction tree:

Goal: P (Possible Pneumonia)

| Rule 3

AND: C ∧ B

/ \\

C (Cough) B (Breathlessness)

True True

\\ /

P proved

Conclusion: Backward Chaining proves Possible Pneumonia because Cough and Breathlessness are both known facts. This is a preliminary rule-based result, not a confirmed medical diagnosis.

Case Study 2 – Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]

II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]

III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA,

Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

The traffic controller uses First-Order Logic (FOL), which represents objects, properties, and relations among objects.

(a) Unification of Rule 1

Rule 1: ∀x [Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)]. Facts: Vehicle(Ambulance) and EmergencyType(Ambulance).

Match Vehicle(x) with Vehicle(Ambulance). The substitution is θ_1 = {x/Ambulance}. Match EmergencyType(x) with EmergencyType(Ambulance); the same substitution succeeds, θ_2 = {x/Ambulance}.

MGU: θ = {x/Ambulance}

After substitution, the rule becomes: Vehicle(Ambulance) ∧ EmergencyType(Ambulance) → ClearPath(Ambulance). Since both premises are facts, derive ClearPath(Ambulance).

(b) Resolution proof of Proceed(CarA)

Resolution refutation proves a goal by adding its negation to the knowledge base and deriving the empty clause □. First convert the relevant rules to CNF.

Rule / fact	CNF clause(s)
Rule 1	¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)
Rule 2, part 1	¬ClearPath(x) ∨ GreenSignal(x)
Rule 2, part 2	¬ClearPath(x) ∨ ¬RedSignal(x)
Rule 3	¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)
Facts	Vehicle(Ambulance); EmergencyType(Ambulance); Vehicle(CarA); Behind(CarA,Ambulance)

Inference 1: Instantiate Rule 1 with x = Ambulance. Resolve ¬Vehicle(Ambulance) ∨ ¬EmergencyType(Ambulance) ∨ ClearPath(Ambulance) with Vehicle(Ambulance), then with EmergencyType(Ambulance), deriving ClearPath(Ambulance).

ClearPath(Ambulance)

Inference 2: Resolve ¬ClearPath(Ambulance) ∨ GreenSignal(Ambulance) with ClearPath(Ambulance).

GreenSignal(Ambulance)

Inference 3: Instantiate Rule 3 with x = CarA and y = Ambulance. Resolve its clause with Vehicle(CarA), Behind(CarA,Ambulance), and GreenSignal(Ambulance).

Proceed(CarA)

Inference 4: Add the negated goal ¬Proceed(CarA). Resolve it with Proceed(CarA) to obtain the empty clause.

Proceed(CarA), ¬Proceed(CarA) ⟹ □

Therefore, KB ⊨ Proceed(CarA). This result follows from the supplied logical rules; real traffic operation would additionally require safety constraints and reliable sensor/control information.

(c) Two simultaneous emergency vehicles

The given knowledge base can infer ClearPath and GreenSignal for each vehicle that satisfies Vehicle(x) and EmergencyType(x). However, it does not specify how to handle two emergency vehicles whose routes conflict or approach the same intersection simultaneously.

Additional knowledge is needed:

Required addition	Reason
ConflictingRoute(x,y)	Identifies routes that cannot safely be used at the same time.

IntersectionOccupied(i)	Tracks whether an intersection is already occupied.
EmergencyPriority(x,level)	Provides a defined policy for deciding which emergency vehicle goes first.
SignalConflict constraint	Prevents incompatible traffic directions from receiving green signals simultaneously.
Tie-breaking rule	Handles equal-priority requests consistently.

A possible simplified rule is ConflictingRoute(x,y) ∧ Proceed(x) → ¬Proceed(y). It must be paired with a clear conflict-resolution policy and applied only where routes truly conflict. Without these additional rules and facts, the KB is not sufficient to guarantee safe handling of simultaneous emergencies.

Case Study 3 – Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded

2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod

3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk

4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

(a) Convert P1, P2, P3 and facts into CNF

Symbols: S = SoilDry; I = IrrigationNeeded; W = CropWheat; D = ApplyDripMethod; R = CropAtRisk.

P1: S → I

Eliminate implication using A → B ≡ ¬A ∨ B:

¬S ∨ I

P1 in CNF: ¬S ∨ I.

P2: I ∧ W → D

Eliminate implication:

$$\neg(I \wedge W) \vee D$$

Move negation inward using De Morgan's law:

$$(\neg I \vee \neg W) \vee D$$

P2 in CNF:

$$\neg I \vee \neg W \vee D$$

P3: $\neg D \wedge W \rightarrow R$

Eliminate implication:

$$\neg(\neg D \wedge W) \vee R$$

Move negation inward using De Morgan's law:

$$(D \vee \neg W) \vee R$$

P3 in CNF:

$$D \vee \neg W \vee R$$

Facts: SoilDry = True and CropWheat = True. Their CNF clauses are S and W.

Complete CNF knowledge base:

$$KB = \{\neg S \vee I, \neg I \vee \neg W \vee D, D \vee \neg W \vee R, S, W\}$$

(b) Resolution Refutation proof of ApplyDripMethod

Goal: D. Negate the goal and add ¬D to the CNF knowledge base.

Clause	Expression
C1	¬S ∨ I
C2	¬I ∨ ¬W ∨ D
C3	D ∨ ¬W ∨ R
C4	S
C5	W
C6	¬D (negated goal)

Resolution 1: Resolve C1 with C4 (S). The complementary S and ¬S cancel, deriving C7 = I.

$$(\neg S \vee I), S \implies I$$

Resolution 2: Resolve C2 with C7 (I). Cancel I and ¬I, deriving C8 = ¬W ∨ D.

$$(\neg I \vee \neg W \vee D), I \implies \neg W \vee D$$

Resolution 3: Resolve C8 with C5 (W). Cancel W and ¬W, deriving C9 = D.

$$(\neg W \vee D), W \implies D$$

Resolution 4: Resolve C9 with C6 (¬D).

$$\mathbf{D, \neg D \implies \square}$$

The empty clause □ proves that the negated goal is inconsistent with the KB. Therefore KB ⊨ ApplyDripMethod.

Resolution proof tree (text form):

$$\mathbf{C1 + C4 \rightarrow C7: I}$$

$$\mathbf{C2 + C7 \rightarrow C8: \neg W \vee D}$$

$$\mathbf{C8 + C5 \rightarrow C9: D}$$

$$\mathbf{C9 + C6 \rightarrow \square}$$

(c) Remove SoilDry from the KB

If fact S is removed, the remaining facts include W, while the rule S → I remains in the KB. Since S is no longer known to be true, Forward Chaining cannot use P1 to derive I.

Without I, P2 cannot derive D (ApplyDripMethod) from the given facts. The rule P3 is ¬D ∧ W → R. Because W is true but ¬D is not established as a fact, CropAtRisk cannot be derived by Forward Chaining either.

Important logical distinction: failure to prove D does not prove ¬D. Therefore, removal of SoilDry alone does not establish CropAtRisk. To derive R using P3, the KB must contain or derive both ¬D and W. If ¬D is explicitly added, then P3 and W permit R.

Conclusion: After removing S, neither IrrigationNeeded nor ApplyDripMethod is derivable from the given facts, and CropAtRisk does not follow unless ¬ApplyDripMethod is separately established.

Case Study 4 – Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The

agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]

2) Breeze [1,1] → Pit is adjacent to cell [1,1]

3) Glitter [x,y] → Gold is at [x,y]

4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new

knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and

Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge

derived in (a) and (b) to justify your conclusion with logical inference.

The agent perceives Stench at [1,2], Breeze at [1,1], and Glitter at [2,2]. The rules provided relate these percepts to nearby hazards, gold, and safety.

(a) Construct the belief state and apply Modus Ponens

Let S_{12} mean Stench[1,2], B_{11} mean Breeze[1,1], G_{22} mean Glitter[2,2]. Let W(x,y) mean Wumpus at a cell, Pit(x,y) mean a pit at a cell, Gold(x,y) mean gold at a cell, and Safe(x,y) mean a cell is safe.

Initial percept facts:

Stench[1,2], Breeze[1,1], Glitter[2,2]

Rule 1: Stench[1,2] → Wumpus is adjacent to [1,2]. By Modus Ponens, derive that the Wumpus is in one of the cells adjacent to [1,2]. The rule does not identify one unique cell.

Rule 2: Breeze[1,1] → Pit is adjacent to [1,1]. By Modus Ponens, derive that a pit is in one of the cells adjacent to [1,1]. The rule does not identify one unique cell.

Rule 3: Glitter[x,y] → Gold[x,y]. Substitute x = 2, y = 2. By Modus Ponens, derive Gold[2,2].

Rule 4: ¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]. This rule can derive Safe for a cell only when both the absence of Wumpus and absence of pit at that specific cell are known. The supplied percepts alone do not establish those negative facts.

Belief state after applying the supplied rules: initial percepts, a Wumpus-adjacency constraint around [1,2], a pit-adjacency constraint around [1,1], and Gold[2,2]. No particular hazard cell or safe cell can be conclusively identified from the rules as stated.

(b) Forward Chaining to identify possible Wumpus and Pit locations

Step 1: Use Stench[1,2]. The Wumpus must be in a cell adjacent to [1,2]. Under standard four-direction grid adjacency, the candidate cells are [1,1], [1,3], [2,2], and [0,2], excluding cells outside the defined world.

Step 2: Use Breeze[1,1]. A pit must be in a cell adjacent to [1,1]. Under standard four-direction grid adjacency, candidates are [1,2], [2,1], [0,1], and [1,0], excluding cells outside the defined world.

Step 3: Use Glitter[2,2]. Derive Gold[2,2]. Glitter does not determine Wumpus or pit location.

Limitation: The question does not specify the complete world boundaries, whether the agent has additional safe-cell facts, or rules that convert each adjacent-hazard conclusion into a formal disjunction. Candidate lists therefore assume a standard orthogonal grid and must be restricted to valid cells in the actual world.

(c) Evaluate the path [1,1] → [1,2] → [2,2]

Cell [1,1] has Breeze, which indicates a pit in a neighboring cell. Cell [1,2] is a candidate pit location because it is adjacent to [1,1]. Cell [1,2] is also a candidate Wumpus location because it is adjacent to the stench at [1,2] only if the world rules allow the current cell to count as adjacent; under standard

grid adjacency, a cell is not adjacent to itself, so that claim must not be made. Instead, the Wumpus candidates are the valid neighbors of [1,2].

The supplied knowledge base does not prove that [1,2] is safe, because no facts establish both ¬Wumpus[1,2] and ¬Pit[1,2]. It also does not prove that [2,2] is safe. Although Gold[2,2] is derived from Glitter[2,2], gold presence does not imply safety.

Therefore, the path cannot be certified safe from the given knowledge base. The agent should gather additional percepts or infer the absence of both hazards at each cell before treating a cell as safe. In a Wumpus World, an unknown cell should not be treated as safe merely because no hazard has yet been confirmed there.

Overall Conclusion

Across the four case studies, logical agents use formal rules and facts to derive conclusions. Forward Chaining starts from facts and derives consequences; Backward Chaining starts from a goal and checks its supporting conditions; Unification matches logical variables with terms; Resolution proves entailment through contradiction; and CNF provides a standard clause form for resolution. The quality of every conclusion depends on the completeness and correctness of the supplied knowledge base. In safety-critical settings such as medicine and traffic control, logical conclusions must be combined with appropriate real-world validation and safety constraints.

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

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A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. $\forall x$: Vehicle(x) $\wedge$ EmergencyType(x) $\rightarrow$ ClearPath(x) [Rule 1]

II. $\forall x$: ClearPath(x) $\rightarrow$ GreenSignal(x) $\wedge$ $\neg$RedSignal(x) [Rule 2]

III. $\forall x \forall y$: Vehicle(x) $\wedge$ Behind(x,y) $\wedge$ GreenSignal(y) $\rightarrow$ Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

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(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

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4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand